Daniel RACOCEANU

BioMedical Image, Information & Data Computing
Professional webpages: https://daniraco.github.io/

SCIENTIFIC PUBLICATIONS / PUBLICATIONS SCIENTIFIQUES

Publications online: https://daniraco.github.io/publications.html

Journals with a reviewing committee

1. Sarbout I., Ounissi M., Milea D., Racoceanu D. (2026) From Synthetic Navigation Data to Real-World Mobility Cues: Reinforcement Learning for Sensory Substitution in Visual Impairment", *Array*, 30:100861. https://www.sciencedirect.com/science/article/pii/S2590005626001840.
2. Ounissi M., Sarbout I., Hugot J.-P., Martinez-Vinson C., Berrebi D., Racoceanu D. (2025). Scalable, trustworthy generative model for virtual multi-staining from H&E whole slide images. *PLoS Comput*. Biol. 21(10): e1013516. https://doi.org/10.1371/journal.pcbi.1013516
3. Zhao, C., Li, J., Zhao, Q., Bai, J., Boluda, S., Delatour, B., Stimmer, L., Racoceanu, D., Jimenez, G., & Fu, G. (2025). *ADNP-15: An open-source histopathological dataset for neuritic plaque segmentation in human brain whole slide images with frequency-domain image enhancement for stain normalization*. *Innovation and Research in Biomedical Engineering* (IRBM), 46(6), Article 100913. https://doi.org/10.1016/j.irbm.2025.100913
4. Sarbout I, Gungor A, Ounissi M, Zaher S , Ptito M, Kupers R, Racoceanu D, Milea D. (2025). Visual Prostheses in the Era of Artificial Intelligence Technology, Eye and Brain, Volume 2025:17 Pages 95—113. DOI : https://doi.org/10.2147/EB.S524322
5. Gungor, A., Sarbout, I., Gilbert, A. L., Hamann, S., Lebranchu, P., Hobeanu, C., Gohier, P., Vignal-Clermont, C., Dumitrascu, O. M., Cohen, S. Y., Lagrèze, W. A., Feltgen, N., van der Heide, F., Lamirel, C., Jonas, J. B., Obadia, M., Racoceanu, D., & Milea, D. (2025). Artificial Intelligence-Based Detection of Central Retinal Artery Occlusion Within 4.5 Hours on Standard Fundus Photographs. *Journal of the American Heart Association*, *14*(13), e041441. https://doi.org/10.1161/JAHA.124.041441.
6. Marin, L.E., Zavaleta-Guzman, D.I., Gutierrez-Garcia, J.I., Racoceanu, D., Casado, F.L. (2025) Prediction of biochemical prostate cancer recurrence from any Gleason score using robust tissue structure and clinically available information. *Discover Oncology,* 16(1), art. no. 128. https://doi.org/10.1007/s12672-025-01896-7.
7. Kumar, P., Lacroix, M., Dupré, P., Arslan, J., Fenou, L., Orsetti, B., Le Cam, L., Racoceanu, D., Radulescu, O. (2024). Deciphering oxygen distribution and hypoxia profiles in the tumor microenvironment: a data-driven mechanistic modeling approach. Phys Med Biol. 2024 Jun 14; 69(12). https://iopscience.iop.org/article/10.1088/1361-6560/ad524a. PMID: 38815610.
8. Ingrassia, L., Boluda, S., Jimenez, G., Kar, A., Racoceanu, D., Delatour, B., Stimmer, L. (2024). Automated deep learning segmentation of neuritic plaques and neurofibrillary tangles in Alzheimer's disease brain sections using a proprietary software, Journal of Neuropathology and Experimental Neurology. https://doi.org/10.1093/jnen/nlae048.
9. Ounissi, M., Latouche, M., Racoceanu, D. (2024). PhagoStat a scalable and interpretable end-to-end framework for efficient quantification of cell phagocytosis in neurodegenerative disease studies, Scientific Reports, https://doi.org/10.1038/s41598-024-56081-7, https://www.nature.com/articles/s41598-024-56081-7.
10. Liu, X., Hu, W., Diao, S., Abera, D. E., Racoceanu, D., Qin, W. (2024). Multi-scale feature fusion for prediction of IDH1 mutations in glioma histopathological images, Computer Methods and Programs in Biomedicine, Elsevier, vol. 248, pp. 108116, doi: https://doi.org/10.1016/j.cmpb.2024.108116
11. Diao S., Tian Y., Hu W., Hou J., Lambo R., Zhang Z., Xie Y., Nie X., Zhang F., Racoceanu D., Qin W. (2022). Weakly supervised framework for cancer region detection of hepatocellular carcinoma in whole-slide pathologic images based on multiscale attention convolutional neural network. *American Journal of Pathology*. 192(3):553-563. doi:10.1016/j.ajpath.2021.11.009.

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13. Jiménez, G., Racoceanu, D. (2019). Deep Learning for Semantic Segmentation versus Classification in Computational Pathology: Application to mitosis analysis in Breast Cancer grading, Front. Bioeng. Biotechnol., 21 June 2019, doi: 10.3389/fbioe.2019.00145.

14. Marin, L., Casado, F., Racoceanu, D. (2019), Classification of prostate cancer based on clinical and omic data using neural networks techniques to improve prognostic power, European Urology Supplements 18(1):e1783; DOI: 10.1016/S1569-9056(19)31292-8.

15. Marin, L., Casado, F., Pinto, J.A., Racoceanu, D., (2019). Classification of prostate cancer based on clinical and omics data using neural networks techniques to improve prognostic power, Journal of Clinical Oncology 37(15_suppl):e16569-e16569; DOI: 10.1200/JCO.2019.37.15_suppl.e16569.

16. Zemouri, R., Zerhouni, N., Racoceanu, D. (2019). Deep Learning in the Biomedical Applications: Recent and Future Status, *Appl. Sci.*, 9(8), 1526; doi:10.3390/app9081526.

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Edited books, symposium proceedings, scientific journals special issues

1. *Special Issue MICCAI 2020*, Purang Abolmaesumi, Anne Martel and Daniel Racoceanu (Eds), *Medical Image Analysis*, sept. 2021, ISSN: 1361-8415.
2. Martel, A.L., Abolmaesumi, P., Stoyanov, D., Mateus, D., Zuluaga, M.A., Zhou, S.K., Racoceanu, D. and Joskowicz, L. (Eds.), *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings (7 volumes / parts), *Springer*: Part I, Part II, Part III, Part IV, Part V, Part VI, Part VII.
3. Racoceanu, D., Hufnagl, P. (Eds), Special Issue - "Scalable and Efficient Imaging Technologies for Digital Pathology" (2017). CMIG - Computerized Medical Imaging and Graphics, Elsevier, vol. 61, pp.1-34.
4. Racoceanu, D., Belhomme, P., (Eds.), Special issue: "Breakthrough technologies in digital pathology", CMIG - Computerized Medical Imaging and Graphics, Elsevier, vol. 2, June 2015, pp.1, DOI: http://dx.doi.org/10.1016/S0895-6111(15)00047-6.
5. Roux, L., Racoceanu, D., (Eds.) Mitosis detection in breast cancer histological images: An ICPR 2012 contest, Journal of Pathology Informatics, special issue, vol. 4, issue 1, 30 May 2013, ISSN: 2153-3539.
6. Loménie, N., Racoceanu, D., Gouaillard, A., (Eds.), Advances in Bio-Imaging: From Physics to Signal Understanding Issues, State-of-the-Art and Challenges, Springer Series: Advances in Intelligent and Soft Computing, vol. 120, 2012, 2012, XXII, 246 p., ISBN 978-3-642-25546-5, January 9, 2012.
7. Leow, W. K., Feng, D., Li, H., Racoceanu, D. (Eds.), SinFra'09 - Singaporean-French IPAL Symposium, Proceedings of the symposium with the selected publications, World Scientific, ISBN: 978-981-4277-55-6(CD) / 981-4277-55-X(CD), 18–20 February 2009, Fusionopolis & Institute for Mathematical Sciences, Singapore.
8. Nakai, T., Racoceanu, D. (Eds.), Special Issue on ONtology and COntext related MEdical image Distributed Intelligent Access, International Symposium of the 26th Annual Meeting of Japanese Society for Medical Imaging Technology (JAMIT), Tsukuba International Convention Center, 20 July 2007, Tsukuba, Japan, Medical Imaging Technology, vol.25 No.5, Nov. 2007.

Book Chapters

1. Popovici, V., Racoceanu, D. (2025). From histopathology images to molecular characterisation of tumours: The artificial intelligence path, in Recent Advances in Histopathology, Vol 27 – Ed. Fred T Bosman and Dr. Ivan Damjanov. Jaypee Brothers Medical Publishers. [DOI]
2. Jiménez, G., Racoceanu, D. (2023). Computational Pathology for Brain Disorders. In: Colliot, O. (eds) Machine Learning for Brain Disorders. Neuromethods, vol 197, pp. 533-572 (40 pages), Humana Press, New York, NY. [DOI]
3. Racoceanu D, Ounissi M, Kergosien Y. (2022) Explicabilité en Intelligence Artificielle ; vers une IA Responsable. Instanciation dans le domaine de la santé. Techniques de l'Ingénieur. Published online December 10, 2022. doi:10.51257/a-v1-h5030 – 22 pages.

4. Zemouri, R., Racoceanu, D. (2021). Innovative Deep Learning Approach for Biomedical Data Instantiation and Visualization, in Deep Learning for Biomedical Data Analysis: Techniques, Approaches/Application, Springer.
5. Lomenie, N., Racoceanu, D. (2012). Ontology-Enhanced Vision System for New Microscopy Imaging Challenges, Advances in Bio-Imaging: From Physics to Signal Understanding Issues, vol. 120, series Advances in Intelligent and Soft Computing, pp 157-172.
6. Loménie, N., Racoceanu, D., Stamon, G. (2011). Point Set Analysis: An Image Analysis Point of View for Rapid Prototyping Technologies, in Rapid Prototyping / Book 1", Ed. M. Hoque, ISBN 978-953-307-970-7, InTech, July 2011.
7. Teodorescu, R. O., Cretu, V. I., Racoceanu, D. (2013). Parkinson's Disease Diagnosis and Prognosis Using Diffusion Tensor Medical Imaging Features Fusion, "Biomedical Engineering, Trends in Electronics, Communications and Software, Ed. Anthony N. Laskovski, ISBN: 978-953-307-475-7, InTech, Jan. 2011 (3000 downloads @ 01/2013).
8. Su, M. J., Cheng, P. H., Chen, S. J., Yang, C. Y., Yip, P. K., Racoceanu, D., Chen, H. S. (2010). Medical Image Intelligent Access Integrated with Electronic Medical Records System for Brain Degenerative Disease, Data Storage, Book edited by: Florin Balasa, ISBN: 978-953-307-063-6, Publisher: InTech, Apr. 2010, pp. 201-211.
9. Lacoste, C., Chevallet, J. P., Lim, J. H., Hoang, D. L. T., Xiong, W., Racoceanu, D., Teodorescu, R. O., Vuillenemot, N. (2007). Inter-Media Concept-Based Medical Image Indexing and Retrieval with UMLS at IPAL, in Lecture Notes in Computer Science, Evaluation of Multilingual and Multi-modal Information Retrieval, vol. 4730, pp. 694-701.

Conferences Invitee

1. Racoceanu D. (2026), Keynote, The International Society for the Processing and Analysis of Medical Information - SIPAIM 2026, Lima, Peru, 18-20 November 2016.
2. Racoceanu D. (2026), From Pixels to Pathways: AI Reasoning over Multimodal Biomedical Data for Precision Medicine, PFEIZER - 6th MLSIG Meeting Series 2026, 5th of Oct. 2026.
3. Racoceanu D. (2026), EMBL-EBI industrial workshop "AI Reasoning with LLMs: Applications in Drug Discovery and Precision Medicine" workshop, Pfeizer, Boston, USA.
4. Racoceanu D. (2026), AAAI Conference on Artificial Intelligence – AAAI-26, SPARTA workshop — Spatial Reasoning and Therapeutics with AI : From Omics to Imaging, Singapore.
5. Racoceanu D. (2025), Hand in hand with XAI on a biomedical imaging journey, Journées Ouvertes en Biologie, Informatique et Mathématiques (JOBIM), Bordeaux, France.
6. Racoceanu D. (2025). Explainable Artificial Intelligence (XAI): a modern Ariadne's thread in Biomedical Imaging. Supporting Discovery of new knowledge, Frugal environmental respecflul Computational Approaches, Computer Vision for Drug Discovery (CVDD) workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA.
7. Racoceanu, D. (2024). Explainable, Hand in hand with AI on a medical imaging journey, Romanian AI days, Bucharest, Romania.
8. Racoceanu, D. (2024). Virtual staining using trustworthy and scalable pipeline, Invited Speaker, European Congress on Digital Pathology – ECDP 2024, Vilnius, Lithuania.
9. Racoceanu, D. (2024). What would Socrates say about AI? - *Guest of Honor* - Annual Conference of the National Academy of Medical Sciences (NAMS), hosted by Bombay Hospital Institute of Medical Sciences, Mumbai India.
10. Racoceanu, D. (2024). Last advances in AI and eXplainable AI, invited speaker, Soirée de Recherche à *l'Hôpital Fondation Adolphe de Rothschild*, Paris, France.
11. Racoceanu, D. (2024). Qvo vadis homine, AI? invited speaker, CEFIPRA Indian-French workshop on XAI for Brain pathologies, Vijayawada, Inde, 19-21 Feb. 2024.
12. Racoceanu, D. (2023). Tumor heterogeneity, tumor micro-environment analysis, modeling and simulation and sparse 3D data reconstruction in computational histopathology, Keynote speaker, MICCAI workshop on Computational Mathematics Modeling in Cancer Analysis (CMMCA2023), MICCAI - International Conference on Medical Image Computing and Computer-Assisted Intervention, Vancouver, Canda.

13. Racoceanu, D. (2023). Explicabilité en Intelligence Artificielle : challenges et opportunités en santé et en recherche biomédicale, invited speaker, Rencontres DIM C-BRAINS -- Google, « IA, cerveau et cloud computing : explorer les frontières de l'innovation », 13 septembre 2023, Paris, France.
14. Racoceanu, D. (2021). Explainable IA: towards a Responsible AI, *keynote speaker*, *SIPAIM 2021* - 17th International Symposium on Medical Information Processing and Analysis, Campinas, Brazil.
15. Racoceanu, D. (2019). Vessels detection for angiogenesis quantification deep learning & marked point process, ECDP 2019 – 15th European Congress on Digital Pathology, Warwick, UK, 10-13 April, 2019.
16. Racoceanu, D. (2019). Tumor Heterogeneity Analysis, Indexing and Simulation in Computational Pathology, Shenzhen Institute of Advanced Technology (SIAT), Shenzhen, China.
17. Racoceanu, D. (2019). Computational Pathology: A Path Ahead, Café de la Neuroinformatique, Institut du Cerveau et de la Moelle épinière (ICM), Paris, France.
18. Racoceanu, D. (2019). Artificial intelligence for biomedical image analysis applications: lessons learned from personal projects, expertise and contributions, Institut du Cerveau et de la Moelle épinière (ICM), Paris, France, 10 May 2019.
19. Racoceanu, D. (2018). Integrative Computational Histopathology with Latin America. Invited speaker, Computational Pathology Symposium, ECP 2018 - 30th European Congress of Pathology, Bilbao, Spain.
20. Racoceanu, D. (2018). A Simulation Model for Computational Histopathology issued from the study of Tumour Microenvironment. An instantiation in Breast Carcinomas, Invited speaker, 14th European Congress on Digital Pathology, Helsinki, Finland, May 29th – June 1st, 2018.
21. Racoceanu, D. (2018). Artificial Intelligence Applied to Very Large Image Analysis in Medicine (Digital Pathology), Invited Speaker, Día Mundial de las Telecomunicaciones, la Sociedad de la Información, y el Internet, Pontifical Catholic University of Peru.
22. Racoceanu, D. (2017). Integrative Computational Pathology and beyond …, Keynote speaker, Digital Pathology and Image Analysis, San Antonio, USA, November 15-16, 2017.
23. Racoceanu, D. (2017). Integrative Digital Pathology and beyond, Invited Speaker, National University of Colombia, Bogota, Colombia, 9th of May 2017.
24. Racoceanu, D. (2016). Integrative digital pathology: semantics and scalable imaging methods, DIGITALPath Europe 2016, London, 18-19 May 2016.
25. Racoceanu, D. (2016). Are we ready to deal with traceability and massive data in Digital Pathology?, Special session on Biomaker Detection and Discovery in Histopathology Images, 2016 IEEE International Symposium on Biomedical Imaging (IEEE ISBI), Prague, Czech Republic, 13-16 April 2016.
26. Racoceanu, D., Rossi, F. (2016). Terres d'innnovation Singapour, une plate-forme d'innovation et de R&D unique en Asie, Invited Speaker, Congres C.U.R.I.E., 6-9 June 2016, Deauville, France.
27. Racoceanu, D. (2015). Semantics, micro-semiology and scalability in future Digital Patholgy, Digital Pathology Congress, 21-22 September 2015, Kuala Lumpur, Malaysia.
28. Racoceanu, D. (2015). Insights about scalability and big data in Digital Pathology. Recent projects and outcomes, Dublin Pathology 2015. 8th Joint Meeting of the British Division of the IAP and The Pathological Society, June 23rd, 2015, Dublin, Ireland.
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37. Racoceanu, D. (2013). Cognitive virtual microscopy for breast cancer grading in histopathology Whole Slide Image exploration using a symbolic cognitive vision approach, Invited Speaker, Centre de Recherche des Cordeliers, UMR INSERM, 8 Apr. 2013, Paris, France.

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39. Racoceanu, D. (2013). Approches cognitives symboliques et connaissances/formes a priori pour l'imagerie microscopique haut contenu. Perspectives à travers des projets franco-singapouriens, Invited Speaker, LTSI, French National Institute of Health and Medical Research, INSERM UMR, University Rennes 1, 14 Feb. 2013, Rennes, France.

40. Racoceanu, D. (2013). Approches visuelles sémantiques en histopathologie, Invited Speaker, Unité BioTICLA - Equipe IMAGIN' Biologie et Thérapies Innovantes des Cancers Localement Agressifs EA 4656 de l'Université de Caen Basse-Normandie, IFR 146 ICORE Centre de Lutte Contre le Cancer François Baclesse, 12 Feb 2013, Caen, France.

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3. Roux, L., Racoceanu, D., (Eds.) Mitosis detection in breast cancer histological images: An ICPR 2012 contest, Journal of Pathology Informatics, special issue, vol. 4, issue 1, 30 May 2013, ISSN: 2153-3539.

International Communications with Proceedings

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3. Ruyter S., Dorent R., Racoceanu D. (2026). Normalization Bias in Morpho-Transcriptomic Prediction, Medical Image and Deep Learning - MIDL 2026 Short Paper Track, Taipei, Taiwan.

4. Ruyter S., Racoceanu D. (2026). When High Correlation Isn't Enough: Rethinking Evaluation in H&E-Based Gene Expression Prediction, ECDP - European Congress on Digital Pathology, Gratz, Austria

5. Ostermaier J., Ruyter S., Racoceanu D. (2026). Cross-Scale Coherence: Evaluating Biological Consistency in Multimodal Latent Spaces, ECDP - European Congress on Digital Pathology, Austria.

6. Sarbout, I., Bodart, G., Cavallo, D., Decottignies, V., Vignal, C., Racoceanu, D., Gohier, P., Chiquet, C., Milea, D. (2026). *Data-driven structure–function model improves central visual field mapping in NAION*, North American Neuro-Ophthalmology Society - NANOS 2026 Annual Meeting, Boston, US.

7. Gungor, A., Zaher, S., Sarbout, I., Croitoru, S., Huna-Baron, R., Touzani, B., Lamirel, C., Banc, A., Racoceanu, D., Audren, F., Keane, P. A., & Milea, D. (2026). *Prospective AI-based retinal image identification of neuro-ophthalmic disorders in a tertiary emergency department*, North American Neuro-Ophthalmology Society - NANOS 2026 Annual Meeting, Boston, US.

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10. Ruyter S., Ounissi M., Racoceanu D. (2025). Diffusion Models for Morphology-Guided Transcriptomics: A Computational Framework, European Congress on Digital Pathology ECDP 2025, Barcelona, Spain.
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14. Milea, D., Gungor, A., Sarbout, I., Hage, R., Gohier, P., Lebranchu, P., Dumitrascu, O., Vignal-Clermont, C., Chen, J.J., Rattanathamsakul, N., Obadia, M., Racoceanu, D. (2024). Diagnosis of Central Retinal Artery Occlusion within 4.5 hours after Visual Loss, using a Deep Learning Method Applied on Fundus Images, North American Neuro-Ophthalmology Society 50th Annual Meeting – NANOS 2024, March 2nd - 7th, 2024, USA.
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47. Racoceanu, D., Diagnostic et pronostic à l'aide des techniques de l'intelligence artificielle, communication dans le cadre de la journée technique organisée par l'Institut de Productique de Besançon, intitulée : « Les atouts de la e-maintenance pour la performance industrielle », Cegelec S.A., Belfort, France, 19 May 2005.
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52. Zemouri, R., Racoceanu, D., Zerhouni, N. (2022) Application des réseaux de neurones récurrents (RRBF) à la détection dynamique de dégradations, Réunion du GRP – Groupement de Recherche en Productique, groupe de travail ASSF - Automatisation et Systèmes Sûrs de Fonctionnement, Grenoble F.
53. Zemouri, R., Racoceanu, D. et Zerhouni, N. (2021) Evaluation des performances d'un atelier de maintenance de PEUGEOT CITROEN AUTOMOBILES, SITE DE Sochaux - France, Proc. of the Symposium International sur la Maintenance Industrielle, SIMI'2001.

Awards

1. Best Scientific Platform Presentation Prize at the biennial European Neuro-Ophthalmology Society (EUNOS) meeting, Milan, Italy, June 2026. Gungor A, Banc A, Zaher S, Sarbout I, Croitoru S, Huna-Baron R, Touzani B, Lamirel C, Racoceanu D, Audren F, Keane PA, Milea D. "Prospective AI-based Fundus Image Identification of Neuro-Ophthalmic Disorders in a Tertiary Emergency Department".
2. Best Rapid Fire Presentation & Poster Prize at the biennial European Neuro-Ophthalmology Society (EUNOS)meeting, Rotterdam, Netherlands, June 2024. Gungor A, Sarbout I, Hage R, Gohier P, Lebranchu P, Dumitrascu O, Vignal C, Chen J, Rattanathamsakul N, Lagreze WA, Feltgen N, Obadia M, Racoceanu D, Milea D. "Early Detection of Central Retinal Artery Occlusion within 4.5 Hours of Visual Loss: Deep Learning Method Applied on Fundus Photographs".
3. Best paper Award finalist at the IEEE International Conference on Robotics & Automation - IEEE ICRA'2001, pp. 46-51, Seoul, Korea - Zemouri, R., Racoceanu, D., Zerhouni, N. (2001). A Petri nets graphic method of reduction using birth-death processes.

Other production

✓ Collective Scientific Reports

1. Eric Tournié, Jean-Marc Allain, Mathieu Arnoux, Nathalie Bardou, Skandar Basrour, Jean-Marc. Bassat, Christian Bergaud, Xavier Brun, Nathalie Destouches, Aurelian Fatu, David Folio, Antoine Grall, Luca Guidoni, Alexandre Guitton, Pierre-Yves Joubert, Arnaud Landragin, Daniel Racoceanu , Christine Restoin, Gwen Salaün, Roger Serra, Didier Trichet, Rapport d'évaluation de l'unité FEMTO-ST – Franche-Comté Électronique, Mécanique, Thermique et Optiques - Sciences et Technologies, Rapport HCERES 2023. https://hal.science/hal-04505233

✓ Scientific reports, submitted papers

1. Swann Ruyter, Reuben Dorent, Daniel Racoceanu, Normalization Bias in Morpho-Transcriptomic Prediction, 2026. https://hal.science/hal-05719756
2. Mehdi Hamadache, Laura Mouton, David Barriere, Cecile Keller, Carine Chassain, Guilhem Pages, Christophe Delhomme, Annick Bernalier, Jean-Marie Bonny, Daniel Racoceanu. Multimodal integration of data characterizing the evolution of the gutbrain axis during the prodromal phase of Parkinson's disease in a rat model. 2025. https://hal.science/hal-05506190v1

✓ Patents and copyrights

Patent / Software (Title & Reference)	**Details & Role**
Virtual staining – paired images *Device and method for generating n virtual IHC stain images from one H&E image (paired img)* EP 24 305 221.4 (Europe, filed 9 Feb 2024) PCT/EP2025/053151 (World filed 6 Feb 2025) WO2025/168731 (World published 14 Aug 2025)	Field: Computational pathology, virtual immunohistochemistry (IHC) Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: Co-inventor & institutional co-owner
Virtual staining – unpaired images *Device and method for generating n virtual IHC stain images from one H&E image (unpaired img)* EP 24 305 224.8 (Europe, filed 9 Feb 2024) PCT/EP2025/053153 (World filed 6 Feb 2025) WO2025/168729 (World published 14 Aug 2025)	Field: Computational pathology, virtual IHC Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: co-Inventor & institutional co-owner
3D Tracking in Microscopy *Method and system for tracking motion of microscopic objects within a 3-D volume* US 2014/0192178 A1, filed 13 Aug 2012, pub. 10 Jul 2014 (A*STAR Singapore)	Field: Biomedical imaging, 3-D cell tracking Role: co-Inventor (holder A*STAR)
Dynamic Real-Time Monitoring System INPI France 0858450, filed 10 Dec 2008 Reference: INPI ref. 8E50BTFR1-Em@systec (Em@systec France)	Field: Industrial vision, sensor systems Role: Co-inventor (holder Em@systec France - spin-off UFC)
Mitosis Detector CNRS DL 05963-01 (UMI 2955 IPAL, 2013)	Field: Computational pathology, cell event detection Role: co-Author (CNRS copyright)
PDFibAtl@s CNRS DL 04137-01 (UMI 2955, submitted 16 Dec 2010, approved 13 May 2011)	Field: Neuroimaging, MRI-DTI data fusion for Parkinson's disease Role: co-Author (CNRS copyright)
HISTOGRAD CNRS DL 2944-01 (UMI CNRS 2955); APPIDDN.fr.001.190019.000.S.P.2010.000.20900 (14 May 2010)	Field: Digital pathology, breast cancer grading Role: co-Author (CNRS copyright)

✓ Regional, National and International Disseminations

1. Racoceanu D., IA en santé, Polytech Dijon, IESF Bourgogne Franche-Comté, dec. 2026.
2. Racoceanu D., IA en santé – la boite noire de la médecine augmentee, Festival de la transition écologique et numérique, Belfort, dec. 2026.
3. Sarbout, I., Cazenave-coupet, T., Ounissi, M., Milea, D., Racoceanu, D. (2026). City of Light (COL): A City-Scale, Geo-Anchored Urban Simulator with High-Throughput Multi-Sensor Streams, Apidays Paris 2025.

4. Working group of the Inria evaluation committee (2025). Reflections on the Use of Generative AI for Research Professions. https://inria.hal.science/hal-05188001. Réflexions sur l'usage de l'IA générative pour les métiers de la recherche. https://inria.hal.science/hal-05187992.
5. Ilias Sarbout, Dan Milea, Daniel Racoceanu - Deep Learning for Navigation of the Visually Impaired using Synthetic Data from Blind Digital Twins, Journées Scientifiques Inria "Handicap, perte d'autonomie & numérique", 19-20 Nov. 2024, Paris ; France.
6. Racoceanu, D., Singapore: Asia's Leading Location for Biomedical and Cosmetic Industries, Guest Editor's Note - FCCS Focus Magazine, French Chamber of Commerce, Singapore, Jan. 2014, pp. 3.
7. IPAL: Four more years for the French-Singaporean lab IPAL, CNRS international magazine, no 22, 2011.
8. Le laboratoire franco-singapourien poursuit son essor, Journal CNRS, no 256, May 2011.
9. IPAL – success stories showcase (2011) of the SECAS EU project funded by the European Commission under the FP7 ICT program; Eutema, a research consultancy from Austria, coordinates SECAS. Other partners include Optimat (UK), National ICT Australia (NICTA) and Singapore Management Univ. (SMU).
10. IPAL: A Decade-Long Success Story, FCCS Focus Magazine, French Chamber of Commerce, Singapore, a Living Lab, summer 2010, pp. 22,23.
11. Singapore, New Research Hub, Around the world, Horizons, CNRS International Magazine n° 18 July 2010, pp. 34

✓ PhD supervision

1. M. Ilias SARBOUT, Measuring, Preventing, and Compensating for Vision Loss: Machine Learning and Digital Twins under Data Constraints, Sorbonne University, Oct. 2026 (co-dir. Dan Milea).
2. Mme. Ayse GUNGOR, Artificial Intelligence in Neuro-Ophthalmology: From Automated Diagnosis to Large-Scale Retinal Dataset Curation, Sorbonne Université, Sept. 2026 (co-dir. Dan Milea).
3. M. Mehdi OUNISSI, Decoding the Black Box: Enhancing Interpretability and Trust in AI for Biomedical Imaging—A Step Towards Responsible Artificial Intelligence, Sorbonne University, Oct. 2024.
4. M. Gabriel Alexandro JIMENEZ GARAY, Representation Learning and Data-Centric Approaches in Computational Pathology. Instantiation to Alzheimer's Disease, Sorbonne University, Sept. 2024.
5. Ms. Laura MARIN, Integrative Digital Pathology for Personalized Medicine: Population stratification and early biomarkers findings, consolidating and completing the use of Prostate-Specific Antigen and Gleason Score in Prostate Cancer, *Univ. Pontificale Catholique du Pérou,* 2023 (co-dir. Fanny Casado).
6. Ms. Oumeima LAIFA, A joint Discriminative-Generative approach for tumour angiogenesis assessment in Computational Pathology, Sorbonne Univ., Sept. 2019, co-dir. Hwee Kuan LEE - BII/A*STAR Singapore.
7. M. Lamine TRAORE, Semantic Modelling of a Histopathology Image Exploration and Analysis Tool, PhD of the University Pierre and Marie Curie, Dec. 2017 (co-dir. Yannick KERGOSIEN).
8. M. Bassem BEN CHEIKH, Graph-based Mathematical Morphology for the Characterization of the Spatial Organization of Histological Structures in High-Content Images: Application to Tumor Microenvironment in Breast Cancer, University Pierre and Marie Curie, Sept 2017.
9. M. Olivier MORERE Deep Learning Compact and Invariant Image Descriptors for Instance Retrieval, PhD of the University Pierre and Marie Curie, June 2016 (co-encad. Antoine VEILLARD - UPMC).
10. Ms. Sreetama BASU (co-dir. A/Prof. Wei Tsang OOI - NUS) Automatic Analysis of Neuronal Morphology: Detection, Modeling and Reconstruction, NUS PhD of the National Univ. Singapore, March 2015.
11. M. Antoine FAGETTE (co-dir. Jean-Yves DUFOUR - Thales) Dense Crowd Analysis, PhD of the University Pierre and Marie Curie, June 2014.
12. M. Stéphane RIGAUD, Analysis-Synthesis Approach for Neurosphere Modelisation Under Phase-Contrast Microscopy, PhD of the University Pierre and Marie Curie, March 2014.
13. M. Humayun IRSHAD, Automated Mitosis Detection in Color and Multi-spectral High-Content Images in Histopathology: Application to Breast Cancer Grading in Digital Pathology, University Joseph Fourier – Grenoble 1, Jan 2014.
14. M. Antoine VEILLARD, Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines, PhD of the National University of Singapore, Dec. 2012.
15. Ms. Roxana Oana TEODORESCU, Parkinson's Disease Prognosis using Diffusion Tensor Imaging Features Fusion / Pronostic de la maladie de Parkinson basé sur la fusion des caractéristiques d'Images par

Résonance Magnétique de Diffusion, joint PhD of the University of Franche-Comté and Politehnica University of Timisoara, Apr. 2011.

16. Ms. Adina Eunice TUTAC (ép. BRANICI), Formal Representation and Reasoning for Microscopic Medical Image-Based Prognosis. Application to Breast Cancer Grading / Représentation et Raisonnement Formels pour le Pronostic basé sur l'Imagerie Médicale Microscopique. Application à la Graduation du Cancer du Sein, joint PhD University of Franche-Comté and Politehnica University of Timisoara, Oct. 2010.
17. Ms. Eugenia MINCA, Surveillance des systèmes de production en utilisant les réseaux de Petri flous. Application à la E-maintenance des systèmes flexibles de production, joint PhD University of Franche-Comté and Univ. Valahia Targoviste, Sept. 2004.
18. M. Nicolas PALLUAT, Méthodologie de surveillance dynamique à l'aide des réseaux neuro-flous temporels, University of Franche-Comté, Jan. 2006.
19. M. Ryad ZEMOURI, Contribution à la surveillance des systèmes à l'aide des réseaux de neurones dynamiques : Application à la e-maintenance, University Franche-Comté, Nov. 2003.

✓ **Habilitation to Supervise Research (HDR – Habilitations à Diriger des Recherches)**

Loménie, N. Habilitation (Habilitation à Diriger des Recherches), Structural and Geometric Analysis of Radiometric Images by Visual Perception Modeling with application to digital pathology, 11th of June 2013, Paris. Jury: Stuart Russel (Univ. Berkeley), Catherine Garbay (CNRS, Univ. Grenoble), Isabelle Bloch (Telecom ParisTech), Laurent Najman (ESIEE & Univ. Marne-la-Vallée), Valérie Gouet-Brunet (Inst. Géographique National), Olivier Lézoray (Univ. Caen), Laurent Wendling (Univ. Paris Descartes), Racoceanu, D. – co-garant (Univ. Pierre Marie Curie), Georges Stamon - garant (Univ. Paris Descartes).

✓ **International Conferences, Symposiums, Special Sessions and Workshops organization**

12. MICCAI 2022, Singapore, Organization Committee (2800 participants)
13. MICCAI 2021, Strasbourg, France, Organization Committee (2700 participants)
14. MICCAI 2020, Lima, Peru – Latin America, **General Chair** (2600 participants)
15. ECDP 2018, European Conf. of Digital Pathology, Helsinki, Finland, Scientific committee.
16. First University of Waterloo - Sorbonne University seminar: " Crystalizing Transdisciplinary innovation in Health Engineering ", 9-11 May, 2016.
17. MITOS & ATYPIA @ ICPR 2014, International Medical Benchmark entitled: Mitosis Detection and Atypia assessment in Breast Cancer Histological Images, in the framework of ICPR 2014, 22nd International Conf. Pattern Recognition, 24-28 Aug 2014, Stockholm, Sweden.
18. ECDP 2014, European Conference of Digital Pathology, Scientific and Organization Committees, 18-20 June 2014, Paris.
19. IJCNN 2013 Special Session: "Computational Intelligence in Bioimage Informatics" (IJCNN-CIBII2013), International Joint Conference on Neural Newtorks, Dallas, TX, USA, August 4-9, 2013
20. SinFra 2012 - Singaporean-French IPAL Symposium, 15-16 October 2012, University Pierre and Marie Curie and Institut Mines-Telecom, Paris - Scientific invited Sessions, Racoceanu, D., General Chair.
21. MITOS @ ICPR 2012, International Medical Benchmark entitled: Mitosis Detection in Breast Cancer Histological Images, initiative of IPAL in collaboration with the SME TRIBVN (Châtillon, France), Hôpital de la Pitié-Salpêtrière (Paris, France) and Ohio State University (US), in the framework of ICPR 2012, 21st International Conference on Pattern Recognition, 11-15 Nov., 2012, Tsukuba, Japan.
22. IJCNN 2011 Special Session "Consciousness-driven vision: toward a breakthrough in bio-inspired computer vision", 31 July – 5 Aug 2011, US.
23. SFBI 2011 – Singaporean - French BioImaging Seminar 2011, 24-25 February. 2011, Biopolis, Singapore (Keynote – Mathias Fink, ESPC, Paris).
24. ICT'2010, EU-ASEAN Cooperation opportunities, Networking Session jointly organized by IPAL and SEACOOP, Bruxelles, Belgium, 27-29 Sept. 2010.
25. Image Understanding - an important step toward Visual Cognition, 2010 IEEE World Congress on Computional Intelligence (WCCI 2010), Special Session, Chairs: Racoceanu, D. and Huang, C. H., 18-23 July, 2010, Barcelona, Spain.
26. SinFra'09, Singaporean-French IPAL Symposium, 18–20 Feb. 2009, Singapore, - Tutorials & Scientific Sessions, Racoceanu, D., Lim, J. H., Leow, W. K., General Chairs (Keynote Joseph Sifakis, 2007 ACM Turing award winner)

27. "Ontology and Context Related Medical Image Distributed Intelligent Access" - ONCO-MEDIA special session at the JAMIT (Japanese Society of Medical Imaging Technology) Annual Meeting, 20th of July 2007, Tsukuba, Japan, chairs: Nakai, T. & Racoceanu, D.
28. Special Session - Medical Image Retrieval - AIRS 2006 – Asian Information Retrieval Symposium, 18 Oct. 2006, Singapore, General Chair - Racoceanu, D., publications issued in Lecture Notes for Computer Science (LNCS)
29. SFBI'06, 1st Singaporean-French Biomedical Imaging workshop, 12-13 October 2006, Biopolis, Singapore, Organizing Committee Chairs: Racoceanu, D., Benoît-Cattin, H., organized with SBIC (Singapore BioImaging Consortium) and CREATIS/INSA Lyon.